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Growth and characterization of mid-far infrared optical material CdSe crystal

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Abstract: We investigated the decomposition conditions of CdSe crystal with the aid of thermodynamics analysis. We chose the optimized growth parameters (1 MPa, 1250°C) based on the numerical simulation, and obtained single crystal boule with 28 mm in diameter and 70 mm in length. Preliminary tests showed that the crystal had fine homogeneity in composition ($CdSe_{0.975}$), high infrared transmission (68-70%, 2.0-16 μm) and large laser induced damage threshold value (LIDT, 56 MW/cm^2). This crystal may be a good candidate for the 8-12 μm mid-far nonlinear optical (NLO) material.

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1. Introduction

Lasers with 8-12 μm mid-far infrared band have many military and civilian applications, involving spectroscopy, infrared countermeasures and atmospheric detection [1]. Though the mature technology of CO_2 laser can output stationary wavelength in range 9.2-10.8 μm,it is unable to cover continuous or spectral regions in the band of 8-9 or 11-12 μm. In order to obtain a wide or a tunable wavelength range, nonlinear optical frequency conversion is the most effective and promising approach. Here, the nonlinear optical (NLO) crystals with wide range of transparency, such as CdSe, $ZnGeP_2$, GaSe, are of great importance [2-3].

Of the-above crystals, CdSe [4] has attracted much more attraction, although its nonlinear optical coefficient is not very large (d_{31} ≈18 pm/V, d_{33} ≈36 pm/V). It has extremely low optical losses (<0.05 cm^{-1}, 1 to 10 μm) and long wave tunability efficiency at λ>8 μm without multi-phonon absorption. Also, CdSe has wide band gap (E_g = 1.7 eV) and transparency range (0.75 to 25 μm), suitable Moh's hardness (3.25) and laser induced damage threshold (LIDT) (≈50MW/cm^2, 20ns, 2.0μm); it can achieve far infrared wavelength even to the range of 8-21 μm, be pumped by 2.09 μm Ho:YAG [5], 2.79 μm Cr, Er:YSGG lasers [6], or be directly used as mid-infrared laser crystal after Cr^{2+}, Fe^{2+} doping [7, 8]. These properties make it one of leading candidates in nonlinear optics devices.

Various methods have been adopted to obtain CdSe single crystal. The temperature gradient solution zoning (TGSZ) technology was described by Burger group [9]. High pressure and vertical zone melting methods were used by The Kolesnivkov group [10]. In addition, chemical vapor transport method (CVT) [11] and solid-state re-crystallization (SSR) [12] were also reported.

In this paper, we report a comparatively simple and effective route to grow optical crystal CdSe with large-size and low defect density. With the help of thermodynamic calculation, we first investigated its decomposition conditions, and then chose the vertical Bridgman growth procedure with moderate safe pressure (1 MPa), and finally obtained a single crystal with 28 mm in diameter and 70 mm in length. We also tested the optical properties of the CdSe crystal, involving the transmittance spectrum and LIDT.

2. The investigation of the decomposition conditions

CdSe crystal has the following decomposition equilibrium under higher temperature (765°C≤ T ≤1300°C) [13].

$$CdSe(s,l) \rightarrow Cd(g) + \frac{1}{2} Se_2(g) \tag{1}$$

The subscripts s, l, and g in Eq. (1) are represented as solid, liquid and gas phases respectively. To simplify the discussion, the gas phase in this system means the decomposed products Cd and Se_2, while the subsequent extra input inert gas are regraded as inert gas.

According to the definition [14], the CdSe thermodynamic equilibrium constant Kx at a fixed temperature T, which expressed in mole fraction and pressure respectively, could be written as following forms:

$$K_x = \frac{x_{Cd} x_{Se_2}^{0.5}}{x_{CdSe}} = \left(\frac{p_{Cd}}{p}\right)\left(\frac{p_{Se_2}}{p}\right)^{0.5} = K_p p^{-\sum_B \nu_B} \tag{2}$$

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$$\left(\frac{\partial \ln K_x}{\partial p}\right)_T = -\frac{\sum_{\mathrm{B}} \upsilon_{\mathrm{B}}}{p} \tag{3}$$

where the x_{Cd}, x_{Se_2}, x_{CdSe} are the mole fraction of Cd, Se_2, CdSe; the p_{Cd}, p_{Se_2}, p are the partial pressure of Cd, Se_2 and the total pressure in the system; and $\sum_B \upsilon_B$ is the difference of gas phase coefficients from decomposed products and reactants that appear in the Equation. The equilibrium composition details are listed in Table 1.

Table 1. The CdSe equilibrium composition varied with the pressure

Title	CdSe	Cd	Se_2	total
Initial amount (mol)	n	0	0	n
Amount at equilibrium α_0, P_0	$n(1-\alpha_0)$	$n\alpha_0$	$0.5n\alpha_0$	$n(1 + 0.5\alpha_0)$
Amount at equilibrium α, P	$n(1-\alpha)$	$n\alpha$	$0.5n\alpha$	$n(1 + 0.5\alpha)$

Here, the α is defined as the degree of CdSe dissociation, n is the initial amount. Keep other experimental parameters invariables, just adjustment the pressure p_0 to p, we would obtain:

$$\frac{K_x^p}{K_x^{p_0}} = \left(\frac{\alpha}{\alpha_0}\right)^{1.5} \times \left(\frac{1+0.5\alpha_0}{1+0.5\alpha}\right)^{0.5} \times \left(\frac{1-\alpha_0}{1-\alpha}\right) \tag{4}$$

From Reference [14], the saturated vapor pressure of CdSe crystal could be gotten:

$$\lg P(bar) = -(11\,088/T) + 7.009 \tag{5}$$

The decomposition α level is calculated to be about 10^{-6}-10^{-3} in the range 765-1300°C. To simplify the discussion, the last two terms in Eq. (4) were ignored. And Equ. 4 may be written as:

$$\ln \frac{K_x^p}{K_x^{p_0}} = \sum_B \upsilon_B \ln \frac{p_0}{p} = 1.5 \ln \frac{\alpha}{\alpha_0} \tag{6}$$

From Eq. (6), the decomposition rate distribution curve under different pressure ratios is shown in Fig. 1 (765°C ≤T ≤ 1300°C). It is easy to see the effect of pressure at a specific temperature: when the pressure increased fivefold, the CdSe decomposition rate drop to 20% and the constant Kx to 9%;, when the pressure increased twenty times, the decomposition rate drop to 5% and the constant Kx to 1%.

Also, the temperature has influence on the decomposition equilibrium of crystal. From the van 't Hoff equations [14]:

$$\ln \frac{K_T^{\vartheta}}{K_{T_0}^{\vartheta}} = -\frac{\Delta_r H_m^{\vartheta}}{R}\left(\frac{1}{T_0} - \frac{1}{T}\right) \tag{7}$$

where $\Delta_r H_m^{\vartheta}, K_T^{\vartheta}$ are the standard reaction enthalpy and equilibrium constant; and R is a constant. Keep other experimental parameters invariable and assume $\Delta_r H_m^{\vartheta}$ as a constant, that decomposition rate will reduce less 5% as the temperature rises per 100°C (Fig. 1, inset).

Therefore, during CdSe crystal growth the increasing pressure can considerably restrain the decomposition, while raising temperature is not a powerful method. Yet, high pressure and high temperature may bring great challenges on the growth crucible, furnace heating and control systems. Therefore, the growth temperature and pressure were chosen to be 1250°C

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and 1 MPa, which is about 20 times of the original decomposed pressure (0.05 × 20 = 1 MPa). This moderate pressure not only reduce the crystal decomposition effectively, but also ensure the safety of the instruments and operation.

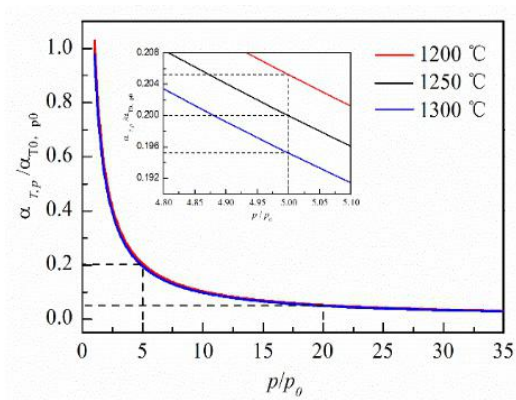

Fig. 1. The diagram of CdSe theoretical decomposition ratio varied with pressure P (inset, the decomposition ratio varied with the temperature T).

3. Experimental procedure

3.1 Preparation of poly-crystalline CdSe

The CdSe poly-materials were prepared in a two temperature zone horizontal furnace with pressure ≤1 MPa. High purity elements Cd, Se (EMei Semiconductor Co. Ltd., Sichuan, China) with 5-9's and 6-9's grade were used as starting materials. The appropriate Cd and Se powders were mixtured and placed in a pyrolytic boron nitride (p-BN) ampoule, then low pressure inert gas was filled in and the ordered lids were fixed tightly. According to the previous report [15], the furnace temperature was first heated to 690°C, then raised to 1270°C and hold at this temperature for about 12 h. After cooling down, 200 g CdSe poly-crystalline could be obtained in one run (Fig. 2(a)).

3.2 Crystal growth

The CdSe crystal is grown in a home-made Bridgman pressure furnace (≤ 2 MPa) with two independent heating zones, where the Si-C (≤1400°C) and Si-Mo (≤1700°C) bars are used as heating source, respectively.

A typical experimental process was described as follows (Fig. 3). First, the p-BN ampoule was filled in approximate CdSe poly-crystalline, nitrogen gas was pumped for 2~3 times with low flow rate, and the ordered lids were tightened. Second, the BN crucible was placed in a Al_2O_3 crucible, a solid BN bar with suitable size was used to minimize the free volume in the internal remaining space, then nitrogen gas was pumped 2~3 times again. Third, the Al_2O_3 crucible was put on a fixed Al_2O_3 supporting frame in the furnace. Subsequently, the furnace cavity was evacuated and injected nitrogen gas for 3 ~5 times. Finally, the furnace inner pressure was kept at about 0.05 MPa.

Then the furnace was set to get a suitable temperature field: the upper one was heated to 1280~1300°C at the rate of 100°C/h, the lower one to 1200~1250°C. Simultaneously, the furnace inner pressure was increased and controlled less than 1 MPa during the whole heating process. The growth process was performed with the temperature invariant for 50 h from the beginning of crystal growth. The temperature gradient with 5-8°C/cm typically at the interface had been chosen, because the low thermal gradient might minimize thermal stresses associated with the anisotropic thermal expansion of CdSe crystal effectively. When the ampoule was cooled slowly to room temperature, a integral CdSe single crystal boule with 28mm in diameter and 70 mm in length was obtained, as shown in Fig. 2(b). Normally, it will

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take 2 weeks to complete the growth process and the growth rate is determined to be 0.4 ~1.2 mm/h. When the crystal was cut with diamond wire saw and roughly polished, CdSe wafers with 6-7 mm in thickness were obtained with no obvious chipping and fracturing, as shown in Fig. 2(c). The CdSe wafers were further cut into elements for laser experiments with about 40mm in length and θ≈71 degree in orientation (Fig. 2(d)).

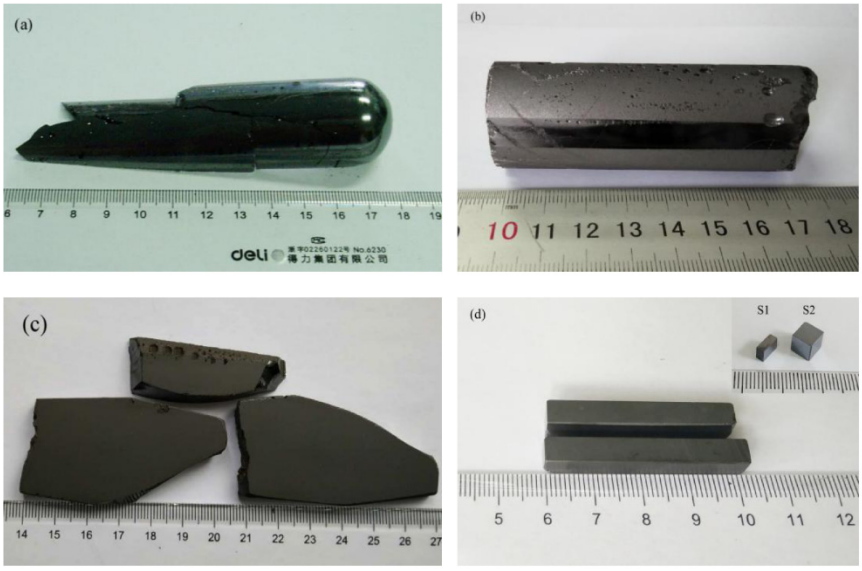

Fig. 2. The photographs of (a) as synthesized CdSe polycrystalline with the weight of about 200 g, (b) as-grown CdSe boules with 28 mm in diameter and 70 mm in length, (c) roughly polished cutting wafers, and (d) some elements about 40mm in length and typical test samples (inset) S1: 4.5 × 6 × 2.5 mm^3, polished face: (100); S2: 5.5 × 6 × 5.5 mm^3, polished face: (002).

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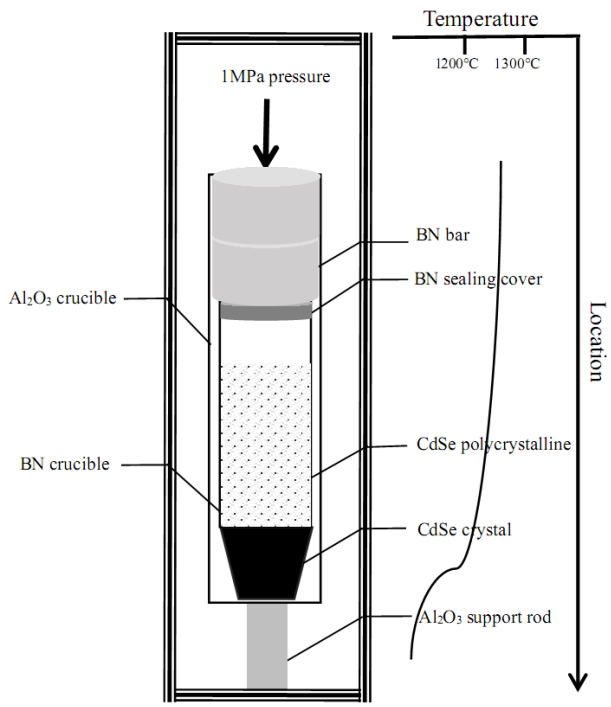

Fig. 3. The schematic diagram of the experiment setup.

3.3 Characterization

The phase and the crystallographic of the products were characterized by X-ray diffraction (XRD) pattern, which was recorded by using a Shimadzu XRD-6000 X-ray diffractometer equipped with CuKα radiation (λ = 0.15406 nm), the scanning rate of 0.05° s^{-1} was applied to record the pattern in the 2θ range of 10-70°.The elemental composition of as-prepared products was analyzed by energy dispersive spectrometer (EDS) attached to a scanning electron microscopy (SEM) (FEI Sirion 200, 15 kV). The visual bulk transparency morphology was observed under a 920nm infrared microscope at room temperature model. Transmittance spectra were measured on Perkin-Elmer Lambda 950 and Bruker Optics Vertex 70 in the range of 700-2500 nm, 4000-480 cm^{-1} respectively. The LIDT experiment was conducted on a home-made 2.09μm Cr, Tm, Ho:YAG laser (pulse duration <50 ns, repetition 1 Hz).

4. Results and discussion

4.1 XRD analysis

Some little blocks from different positions of the crystal were collected and ground into powder. A typical XRD pattern was recorded as shown in Fig. 4(a). The diffraction peaks are labeled and indexed as pure hexagonal phase CdSe. The cell parameters are calculated to be $a = b = 0.4302 \pm 0.002$ nm and $c = 0.704 \pm 0.005$ nm, which are also in agreement with the reported values of $a = b = 0.43$ nm, $c = 0.702$ nm [JCPDS 8-459].

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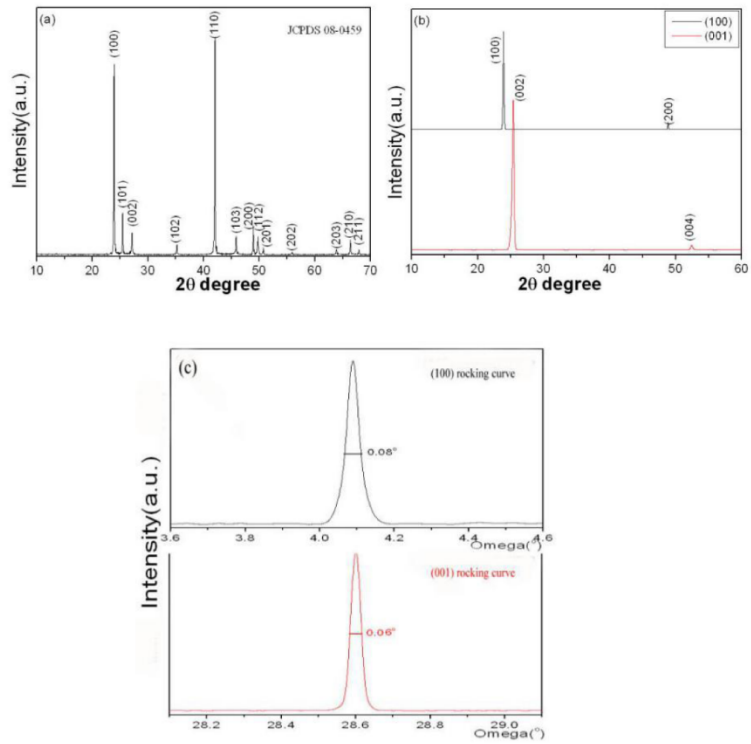

Fig. 4. The as grown CdSe crystal's (a) power XRD pattern, (b) (100) and (002) single crystal diffraction, and (c) corresponding rocking curves.

In order to confirm the crystallinity of the as-grown CdSe crystal, samples S1 (size: 4.5 × 6 × 2.5 mm^3, (100)), S2 (size: 5.5 × 6 × 5.5 mm^3, (002)) were fabricated (Fig. 2(c), inset). The typical XRD patterns of S1 and S2 are shown in Fig. 4(b). No peaks except the ordered diffraction peaks {100} and {002} are observed. Figure 4(c) are the corresponding XRD rocking curves. The intensity of the diffraction peak is high and the shape of the peak has good symmetry with the full widths at half maximum (FWHM) of about 0.08° and 0.06°, which demonstrates that both samples have high crystallization quality.

4.2 Composition examination

Three crystals samples with about 20 mm intervals along the boule were prepared for the composition test, with 9 independent points spreading over an area above 23 mm^2 in each sample. Figures 5(a) and 5(b) are the typical SEM image and EDS pattern. The EDS shows that the weight ratio of Cd to Se is about 0.593: 0.407, corresponding to the atomic ratio of 50.58: 49.42. In all the EDS detection, the atom ratio of Cd to Se varies from 50.22: 49.78 to 50.98: 49.02. These ratio indicates that the CdSe crystal is near the ideal one with the average chemical formula of $CdSe_{0.975}$. The grown condition with moderate pressure really help improve the homogeneity of the crystal in composition.

In fact, the atom ratio is interesting and little different from the theoretical calculation. Based on the three phases equilibrium, the content of Se element should be tiny larger than that of the Cd element. In this work, the Se element loss is still a little larger than we expect, probably because of the high vapor pressure of Se element during the synthesis and growth processions.

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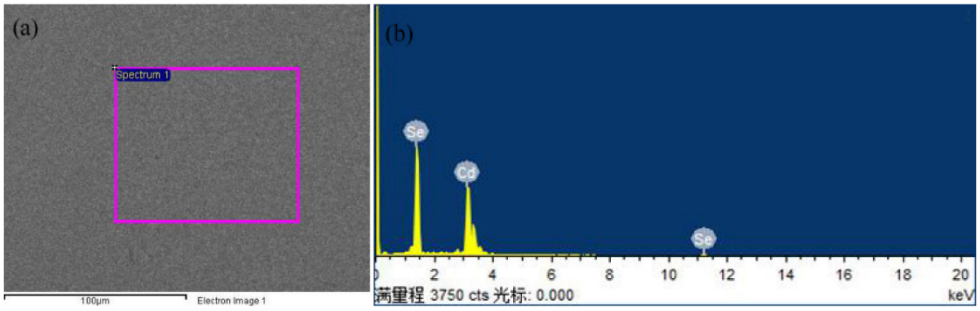

Fig. 5. (a) SEM image of CdSe crystal and (b) the EDS of CdSe revealing its chemical composition of Cd: Se = 50.58: 49.42.

4.3 Optical properties

The short cut-off wavelength of CdSe crystal is about 0.75 μm. Therefore, we cannot judeg the opaque crystal's quality by naked-eyes. Figure 6(a) is a typical image of the 2-mm-thick crystal with the word "CdSe" under it using a 920 nm infrared microscope, the spot size's diameter being about 8 mm. We can see that the crystal is free of voids, twins and phase precipitates. And the word "CdSe" can be seen clearly.

The above-mentioned spectrometers were employed to verify the crystal's optical property in the whole transmission range. The specific wavelength transmittance spectra of Samples S1 and S2 with thickness 2.5 mm and 5.5 mm respectively were recorded and shown in Figs. 6(b) and 6(c). At "0" transmittance level, the fundamental absorption edges are both at 732 nm, corresponding to the energy band-gap 1.69 eV (Fig. 6(b) inset). Also, the results coincide well with the two different instruments in position 2.5 μm (4000 cm^{-1}). And the samples exhibit high transmittance with about 68-70% in the 2.0-16.0 μm wavelength. According to reference [16], the transmittance absorption coefficients values were calculated and shown in Fig. 6(d) with mean values of 0.04-0.05 cm^{-1}. These results reveal that the absorption coefficients are little independent of the thickness, which show that the samples have uniform and fine optical quality.

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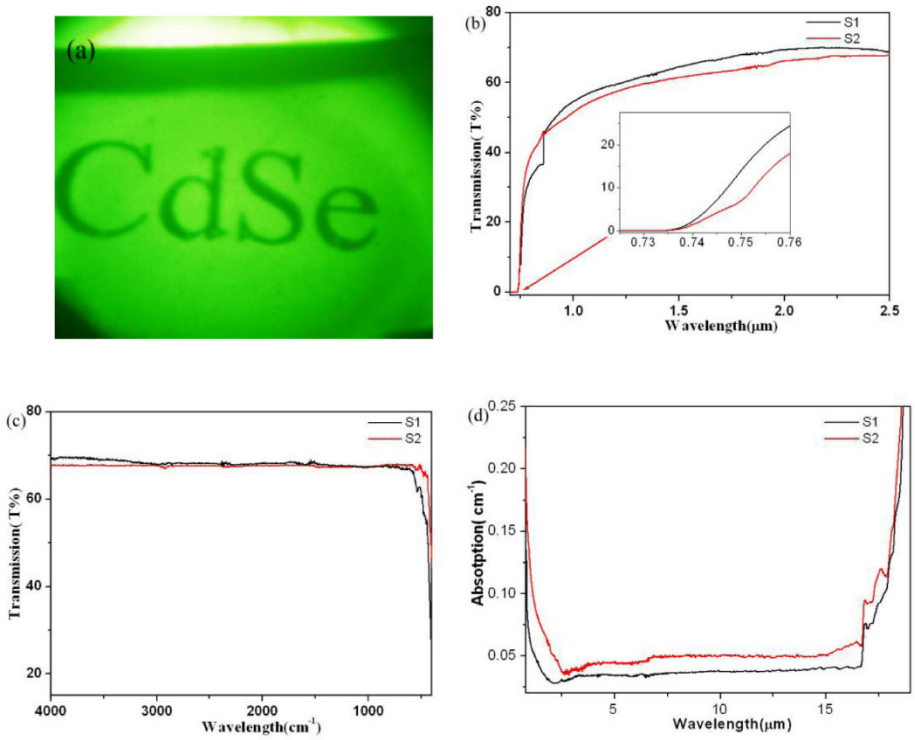

Fig. 6. (a) The bulk transparency image under 920 nm infrared microscope; The transparency spectrum of CdSe samples S1 and S2 with thickness 2.5 and 5.5 mm respectively: (b) in the range of 700-2500 nm (inset: the magnified image of corresponding cutoff wavelength) and (c) in the range of 4000-480 cm^{-1}; and (d) the optical absorption spectra in the 950-17000 nm by calculation.

LIDT is one of the important factors for the evaluation of nonlinear optical crystal quality, especially in the high energy output applications. Here, the standard of 1-on-1 test procedure (ISO11254-1, 2011) was adopted to test the CdSe crystal. A sample with 6 × 5 × 2 mm^3 was prepared in the interaction Type II (θ = 0 degree, Φ random angle). With this processing type, the test result may be close to the real phase-matching operating environment. Then the 6 × 5 section was divided into 16 spots and 4 of them were classified as an individual group. Each spot underwent one irradiation and each group was irradiated with the same energy. Next, a home-made 2.09μm Cr, Tm, Ho:YAG laser with pulse duration <50 ns, repetition 1 Hz, beam diameter 0.8-1.2 mm was used as a laser source. The power of the laser beam increased and marked one group. Simultaneously the spots were observed under an optical microscope. The pulse fluence that manages to make the corresponding damage probability values of 20% was defined as the LIDT. At the same time, a 2mm thick $AgGaSe_2$ crystal with absorption coefficient about 0.03-0.07 cm^{-1} in 2.0-12.0 μm (Fig. 7, inset) was used as the reference with the other experimental conditions invariant. The corresponding transmission of the $AgGaSe_2$ crystal was shown in Fig. 7.

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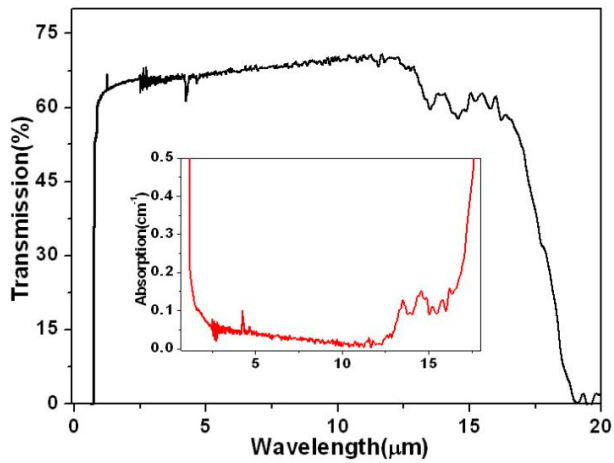

Fig. 7. The transparency spectrum of $AgGaSe_2$ in the range of 700-20000 nm (inset: the corresponding optical absorption spectrum in the 800-17500 nm by calculation).

The experimental results show that the surface LIDT value of CdSe crystal was about 56 MW/cm^2 (2.09 μm, 46 ns, 1Hz), while that of the referential $AgGaSe_2$ was determined to be about 18 MW/cm^2. Such a large value could be anticipated with anti-reflection (AR) coating. Although it has been widely accepted that a wide band gap supports a strong resistance to high-power laser irradiation and the value of CdSe (1.69 eV) and $AgGaSe_2$ (1.70eV) are almost the same, the CdSe crystal exhibits an encouraging result. The phenomenon may be explained as following: Ag-based compounds, such as $AgGaSe_2$, probably show photo darkening phenomenon, while the Cd-based compounds avoided it and showed their real LIDT values.

5. Conclusions

In summary, the homogeneities bulk CdSe crystal with size of Φ 28 × 70 mm^3 was grown by vertical Bridgman method under 1 MPa pressure. Preliminary test showed that the FWHM of crystal (100) (002) peaks are 0.08° and 0.06°. The transmittance is about 68-70% with mean absorption 0.04-0.05 cm^{-1} in the range 2.0-16.0 μm, the LIDT was measured to be 56 MW/cm^2 (2.09 μm, 46 ns, 3Hz), about three times of that of high quality $AgGaSe_2$ crystal. It may be a good NLO candidate for mid-far (8-12 μm) infrared application.

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